

Fig. 6: An actual-size, single-side PCB for the namaste greeting robot

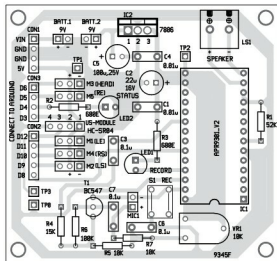


Fig. 7: Component layout for the PCB

objects or environment. Arduino Uno is a board based on ATmega328 microcontroller. It comprises 14 digital input/output pins, six analogue inputs, a USB connection for programming the on-board microcontroller, power jack, an ICSP header and a reset button. It is operated with a 16MHz crystal

oscillator and contains everything needed to support the microcontroller. It is very easy to use as the user simply needs to connect it to a computer with a USB cable, or power it with an AC-to-DC adaptor or battery to get started. The microcontroller on the board is programmed using Arduino programming language and Arduino development environment.

Pin 4 of Board1 is connected to LED2 which glows when the robot detects someone and greets with 'namaste.' Pin 8 is connected to pin 23 of IC1 to play the recorded message. This pin goes low when robot detects someone through the ultrasonic module.

The circuit uses two 9V batteries: one to power the Arduino Board (Board1) and the other to power the recording and playback circuit as shown in Fig. 2. The 5V regulated supply for servos and ultrasonic module is provided by the Arduino board itself. A glowing LED on Board1 indicates the presence of power supply.

Software

The software for the robot is written in Arduino programming language. The Arduino Uno board is programmed using Arduino IDE software. Atmega328 on Arduino Uno board comes with a boot loader that allows you to upload new code to it without the use

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on source.efymag.com

of an external hardware programmer. It communicates using the STK500 protocol. You can also bypass the boot loader and program the microcontroller through in-circuit serial programming (ICSP) header, but using boot loader programming is quick and easy.

Select the correct board from 'Tools→Board' menu in Arduino IDE and burn the program (sketch) through standard USB port in the computer.

Construction and testing

An actual-size, single-side PCB for the *namaste* robot circuit is shown in Fig. 6 and its component layout in Fig. 7. Assemble the circuit on the PCB to save time and minimise any assembly errors. Program the Arduino board with software provided in EFY DVD and at source.eftymag.com, and connect it to the assembled PCB as shown in Fig. 2.

Power up the record and playback system with a 9V battery connected at BATT.2. Now press and hold switch S1 to record the message. Connect all the servo motors and ultrasonic module and power up the Arduino section as well by connecting another 9V battery at BATT.1. Now the robot is ready to work.

To test the circuit for proper functioning, verify the test points as mentioned in test points table. ●

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